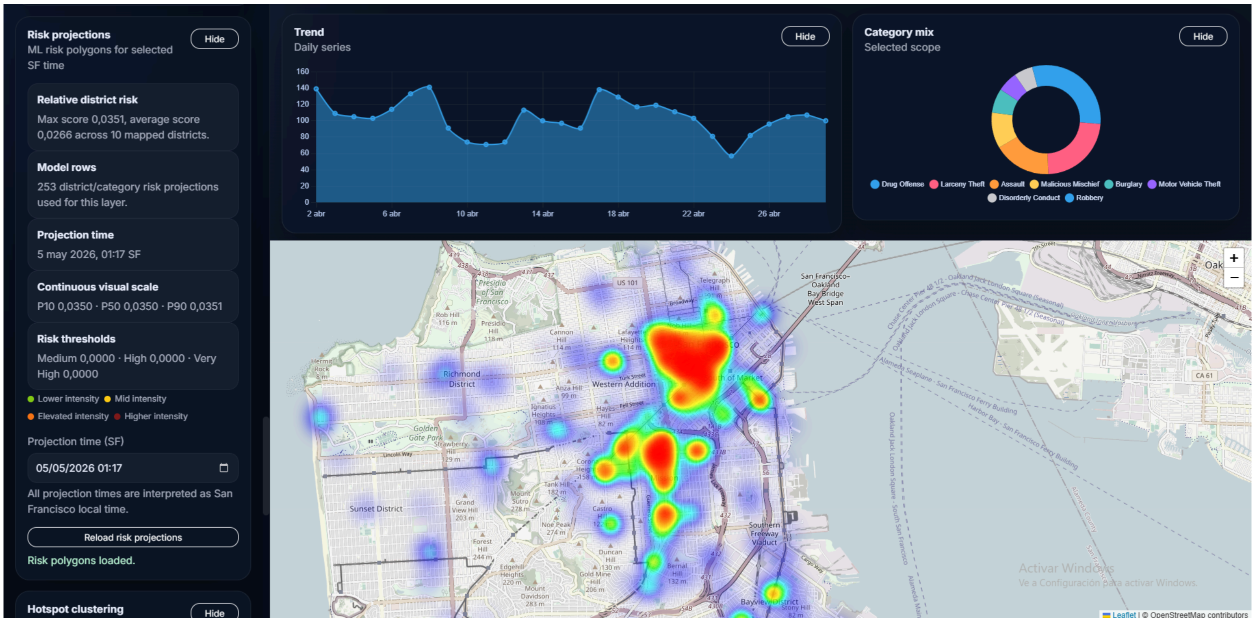
CI San Francisco



Author: Maximiliano Alarcón
Tech Stack: Flask, JS, PostgreSQL, S3, Scikit Learn
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Status: Production-ready
Region: San Francisco Bay Area

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1. Product Overview

CI SF (Crime Intelligence San Francisco) is a data-driven web application designed to analyze, visualize, and interpret crime activity across San Francisco in a clear and actionable way.

The platform transforms raw incident data into:

- real-time insights
- spatial risk awareness
- predictive signals

It enables users to:

- understand how crime is distributed across the city
- detect high-density hotspot areas
- evaluate district-level risk exposure
- explore short-term projections of future activity

Unlike traditional dashboards, CI SF does not simply display data — it provides contextual intelligence.
By combining data engineering, spatial analysis, and lightweight machine learning, the system highlights patterns, exposes hidden structure in the data, and preserves critical context such as isolated incidents that would otherwise be ignored.

2. Key Features

Interactive Crime Map

- Incident-level visualization
- Density heatmaps and marker layers
- Risk-based color encoding

Hotspot Clustering (DBSCAN)

- Detection of high-density crime zones
- Visualization using dynamic radius polygons
- Isolated incidents rendered as contextual points

Risk Signals & KPIs

- Open/Active case ratio
- Online reporting share
- Incident trends over time

District Intelligence

- Ranking by incident pressure
- Category distribution per district
- Comparative analysis across zones

Forecasting Layers

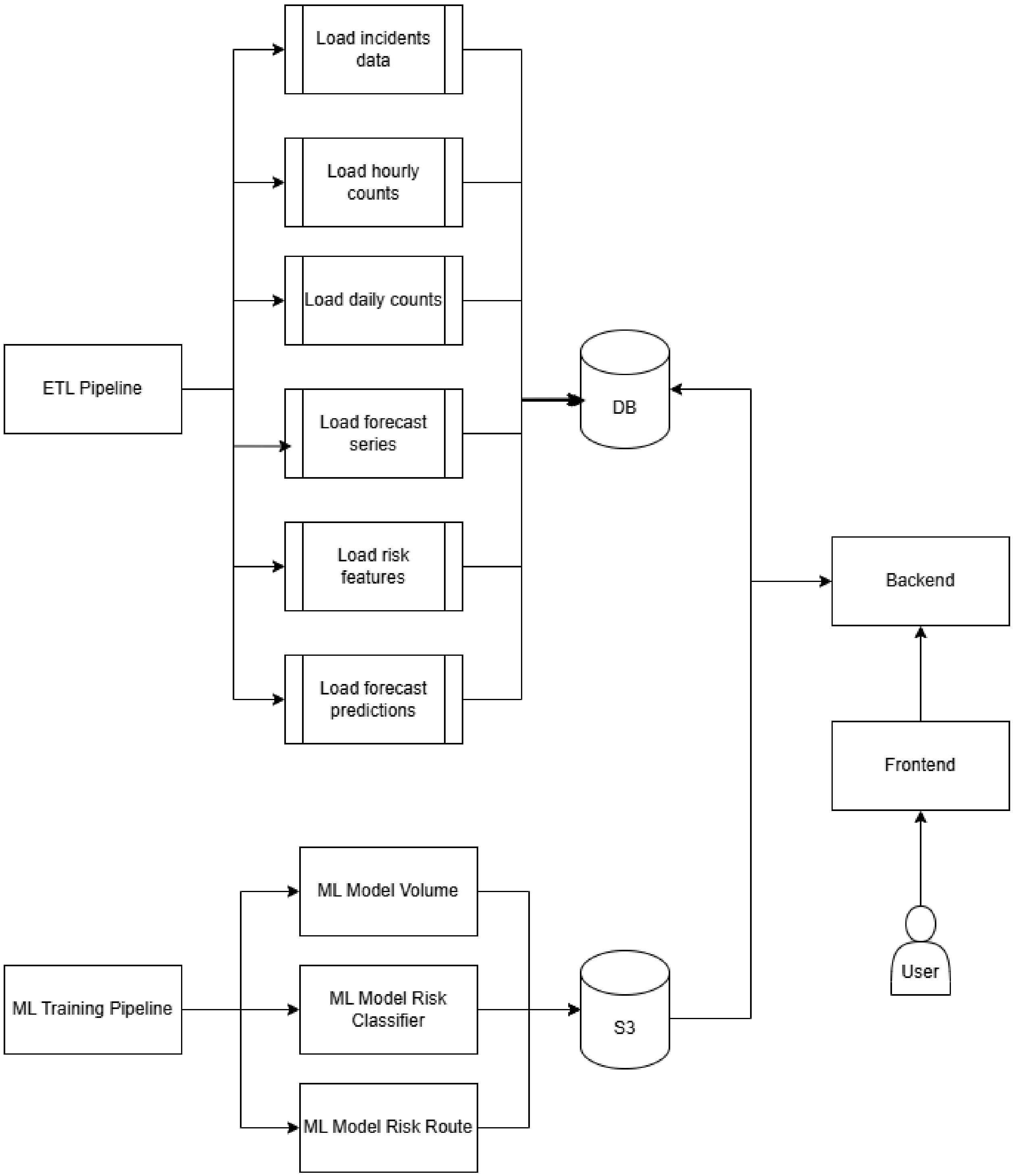
- Volume prediction (next-hour incidents)
- Risk projection (continuous intensity scale)

Advanced Filtering

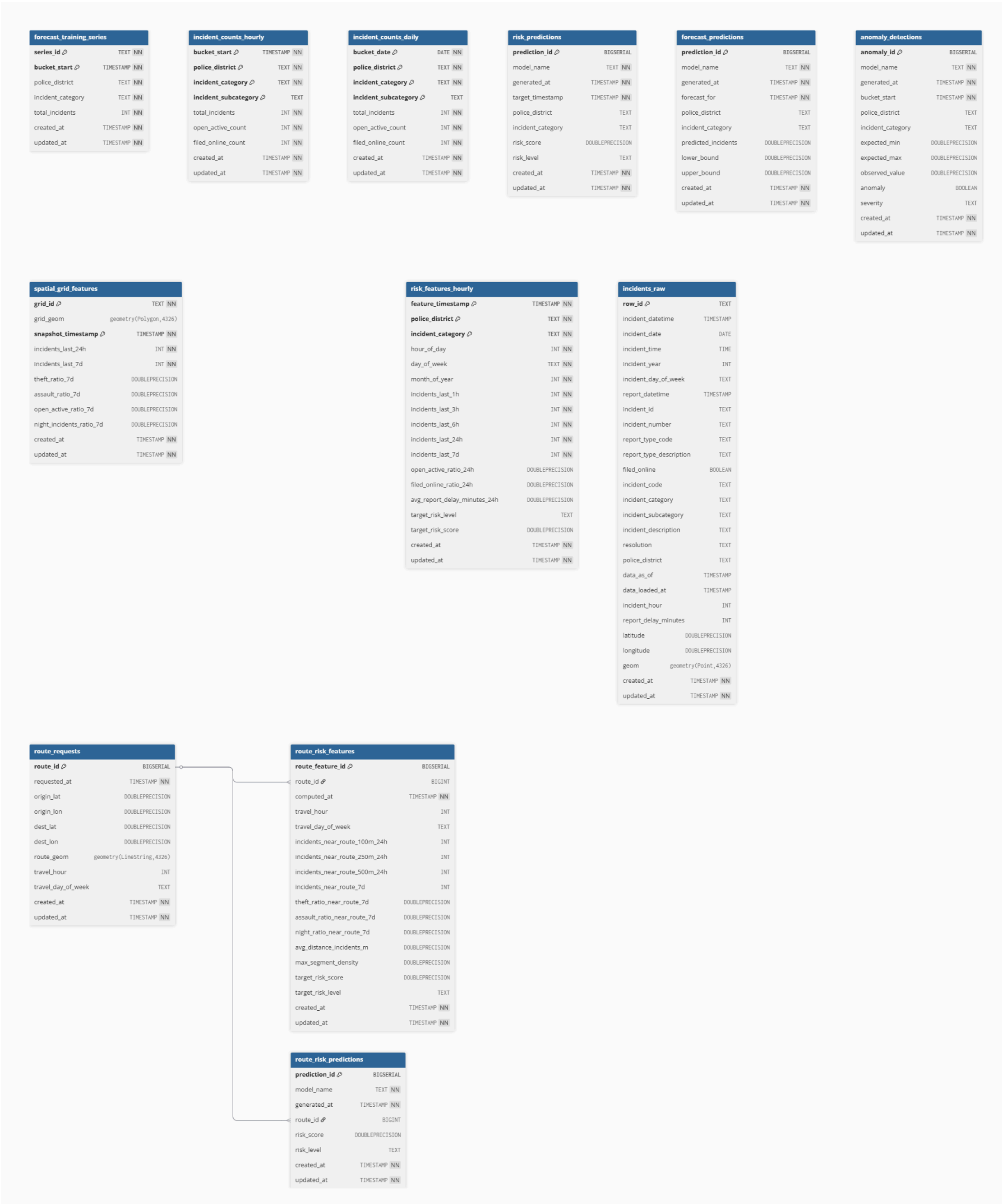
- Time windows (24h, 7d, 30d)
- District-level filtering
- Category-specific analysis

3. Architecture

Overview



Database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4. Core Engineering

CI SF combines spatial analysis, time-series modeling, and real-time visualization into a unified system. The following components represent the core engineering decisions behind the platform.

Spatial Clustering (Hotspot Detection)

Hotspots are generated using a DBSCAN-based clustering approach over geospatial incident data.

Key characteristics:

- Haversine distance for accurate geographic proximity
- Dynamic clustering per incident category
- Density-based grouping to detect high-activity zones

A critical design decision was the treatment of noise (isolated points):

- Instead of discarding them, they are explicitly preserved
- Rendered as contextual markers to avoid losing spatial information

This ensures the system captures both:

- high-density patterns (clusters)
- low-density signals (isolated incidents)

Risk Modeling (Heuristic + ML Hybrid)

Risk estimation combines engineered features with predictive modeling to produce actionable outputs.

The system integrates:

- historical incident volume (multi-window aggregation)
- resolution status (open/active ratio)
- reporting behavior (filed online)
- temporal patterns (hour, day, seasonality)

These inputs are transformed into:

- a normalized risk score (0–1)
- a categorical risk level (Low → Very High)

Additionally, forecast signals and anomaly detection contribute contextual awareness, enabling the model to react to short-term changes rather than only long-term patterns.

Time-Series & Prediction Pipeline

The platform includes lightweight forecasting and anomaly detection models:

- Forecasting estimates expected incident volume per district/category
- Anomaly detection compares observed vs expected behavior
- Risk scoring integrates both historical features and predictive signals

This layered approach allows the system to:

- anticipate future activity
- detect unusual spikes
- adjust risk dynamically

Visualization Engine

The frontend is designed to balance information density with clarity.

Map layers include:

- incident markers (point-level data)
- density heatmaps
- hotspot polygons (clusters)
- prediction overlays

Key design considerations:

- clear layer separation to avoid visual clutter
- consistent color hierarchy across features
- perceptual clarity on desaturated (grayscale) map tiles

This ensures that complex analytical outputs remain interpretable in real time.

5. Data & Pipelines

Data Overview

Table	Description	ETL	Usage / ML model
incidents_raw	Stores raw incident data from the San Francisco Police dataset. Fetches data from Socrata API as incremental ingestion	load_incidents.py	Used by Risk Route model
incident_counts_hourly	Hourly aggregation of incidents.	refresh_hourly_aggregates.py	Used to generate: risk features , forecast series and ML training datasets
incident_counts_daily	Daily aggregation (lighter version of hourly).	refresh_daily_aggregates.py	Mainly for analytics / dashboard
risk_features_hourly	Core feature engineering table for ML. Combinations of hour × district × incident category	build_risk_features.py	Risk classifier model and Volume model
forecast_training_series	Time-series dataset for forecasting models.	build_forecast_series.py	Forecast model (baseline + ML) and anomaly detection
forecast_predictions	Stores predicted incident volume.	build_predictions.py	Used internally in: risk scoring, anomaly detection. Not exposed in UI (yet)
anomaly_detections	Detects abnormal incident behavior.	build_predictions.py	Used for contextual signals. Can enhance alerts / dashboards
risk_predictions	Final risk output used in the UI.	build_predictions.py	Final output of risk pipeline. Consumed by Frontend
route_requests	Stores route queries (for route risk analysis).	None	Created via API requests
route_risk_features	Feature dataset for route-level risk.	None	Derived from route + spatial incident data
route_risk_predictions	Predicted risk for each route or segment.	None	Generated by Route model

Machine learning Overview

Model	Description	Objective	Data Used
Risk (District - Level)	Predicts the overall crime risk for a given district, category, and time window using historical patterns and engineered features.	Estimate risk level (Low → Very High) and risk score for each zone to support decision-making and visualization in the dashboard.	risk_features_hourly, incident_counts_hourly
Volume Forecast	Predicts the expected number of incidents in the next hour based on historical time-series patterns.	Estimate future incident volume to anticipate changes in activity and support downstream models.	risk_features_hourly, incident_counts_hourly
Forecast (baseline seasonal)	Uses historical hourly patterns (same hour/day) to estimate expected incident counts and confidence bounds.	Provide baseline predictions and expected ranges for anomaly detection and risk calibration.	forecast_training_series
Anomaly detection	Compares observed incidents against expected historical ranges to detect unusual activity.	Identify unexpected spikes or drops in crime activity and assign severity levels.	forecast_training_series, incident_counts_hourly
Risk (Route - Level)	Estimates the probability of incidents along a specific route or segment using spatial proximity and transport mode.	Evaluate risk along a path, enabling safer route selection (e.g., walking vs transit).	incidents_raw (spatial queries), generated route geometries

6. API / Interfaces

The backend is implemented as a REST API using Flask, exposing endpoints for data retrieval, analytics, machine learning predictions, and system operations.

The API is organized into four main groups:

- Dashboard APIs (data for UI)
- Machine Learning APIs
- Administrative APIs

Dashboard APIs

These endpoints power the main UI and provide processed, aggregated, and spatial data.

/api/dashboard/filters

Returns available filter values (districts and categories) based on current data.

/api/dashboard/overview

Provides key KPIs:

- total incidents
- open/active ratio

/api/dashboard/trend

Returns time-series data:

- hourly or daily incident trends
- open/active counts

/api/dashboard/district-pressure

Returns top districts ranked by incident volume and pressure.

/api/dashboard/category-mix

Returns category distribution for charts.

/api/dashboard/risk-signals

Provides aggregated signals derived from feature data:

- rolling incident windows
- open ratio

/api/dashboard/map-points

Returns incident-level data for map visualization:

- coordinates
- category
- resolution
- computed risk score

Includes:

- dynamic marker sizing
- risk-based color weighting

/api/dashboard/hotspots

Generates hotspot clusters using DBSCAN:

- cluster polygons
- isolated incidents (noise points)
- density metrics

Machine Learning APIs

These endpoints expose model predictions and derived spatial outputs.

/api/dashboard/ml-volume-polygons

Returns predicted incident volume as GeoJSON polygons.

- Uses volume model
- Supports time projection (target_timestamp)
- Aggregated by district

/api/dashboard/ml-risk-predictions

Returns raw risk predictions:

- per district/category
- includes risk score outputs

/api/dashboard/ml-risk-polygons

Returns spatial risk visualization:

- district-level polygons
- risk scores mapped to geometry

/api/ml/route-risk (POST)

Evaluates risk for specific routes or itineraries.

Input:

- route geometry (legs)
- transport mode

Output:

- risk score per route
- model metrics

Supports:

- integration with routing systems (e.g., GPS SF)

Administrative APIs (Protected)

These endpoints require an admin token and are used for model execution and ETL control.

/admin/ml/volume/model-info

Returns metadata for the volume prediction model.

/admin/ml/volume/predict (POST)

Runs volume predictions on demand.

/admin/ml/risk/model-info

Returns metadata for the risk model.

/admin/ml/risk/predict (POST)

Runs risk predictions on demand.

7. Challenges & Solutions

This project involved several practical challenges related to data quality, spatial analysis, and real-time visualization. The following solutions reflect key engineering decisions made during development.

Data Quality: Missing Coordinates

Problem
A significant portion of incidents lacked latitude/longitude values, making them unusable for map-based analysis.

Solution
A backfill strategy was implemented in the ETL layer to:

- recover missing coordinates when possible
- ensure a consistent dataset for spatial queries

Impact
Improved map coverage and enabled reliable hotspot detection.

Spatial Clustering: DBSCAN Hiding Data (Noise)

Problem
DBSCAN naturally classifies low-density points as noise, which can lead to loss of potentially relevant information.

Solution
Instead of discarding noise:

- isolated incidents are preserved
- rendered as contextual points on the map

Impact
The system captures both:

- high-density clusters (hotspots)
- low-density signals (isolated events)

This was a key product decision to avoid misleading visualizations.

Visualization: Map Clutter and Overlapping Layers

Problem
Displaying raw points, clusters, and heatmaps simultaneously can create visual overload and reduce usability.

Solution
Implemented:

- clear layer separation (points, heatmap, clusters, predictions)
- opacity tuning and size scaling
- selective rendering based on context

Impact
Maintains clarity while preserving high information density.

Data Consistency: Category Fragmentation

Problem
Incident categories from the source dataset are inconsistent and fragmented (e.g., variations of similar categories).

Solution
Introduced a curated category whitelist:

- filters relevant categories
- normalizes analysis across endpoints

Impact
Improved consistency in dashboards, clustering, and ML models.

Performance: Large Dataset Handling

Problem
Rendering and processing large volumes of incident data can impact both backend performance and frontend responsiveness.

Solution
Multiple optimizations were applied:

- point limits for map rendering
- indexed queries in PostgreSQL
- pre-aggregated tables (hourly/daily)

Impact
Ensures responsive UI and scalable query performance.

8. Performance & Scaling

CI SF was designed with performance in mind from the beginning, which prevented major scalability issues during development.

Rather than reacting to bottlenecks, the system incorporates lightweight design decisions that ensure consistent performance across typical workloads.

Backend Efficiency

The backend leverages a combination of strategies to keep queries fast and predictable:

- Pre-aggregated tables (hourly and daily) to avoid expensive real-time computations
- Indexed time-series queries for efficient filtering by time window
- Controlled query scopes using filters (district, category, time range)

These design choices ensure that most endpoints operate on bounded and optimized datasets rather than raw data.

Frontend Rendering

To maintain responsiveness in the UI:

- Map point rendering is limited using a configurable threshold (`MAP_POINT_LIMIT`)
- Layer toggling allows switching between markers, heatmaps, and clusters
- Conditional rendering avoids unnecessary DOM updates

This prevents performance degradation when visualizing large numbers of incidents.

Lightweight Machine Learning

The ML pipeline is designed for fast inference:

- Models use precomputed features (`risk_features_hourly`)
- No heavy real-time training or large model loading is required
- Predictions are generated quickly from structured inputs

This enables real-time interaction without noticeable latency.

Scaling Considerations

Although the system has not faced performance bottlenecks yet, it is structured to scale:

- Clear separation between raw data, aggregates, features, and predictions
- ETL pipelines reduce runtime computation
- API endpoints operate on preprocessed data rather than raw ingestion

9. Future Improvements

The current system provides a solid foundation for spatial analytics and risk modeling. Several enhancements can significantly extend its capabilities and move it closer to a production-grade intelligence platform.

Advanced Machine Learning Models

Current models are lightweight and designed for fast inference. Future iterations could include:

- Gradient boosting models (e.g., XGBoost) for improved predictive accuracy
- Temporal models (e.g., LSTM or sequence-based approaches) to better capture sequential patterns
- Feature importance analysis to improve interpretability

Goal: increase prediction quality while maintaining real-time performance.

Multi-Step Forecasting

The current system predicts only the next time window (e.g., next hour).

Future improvements include:

- multi-step forecasting (e.g., next 6h, 12h, 24h)
- confidence-aware projections over longer horizons

Goal: enable medium-term planning and trend anticipation.

Route-Aware Risk Integration

A key extension is the integration with the GPS SF system:

- combine routing with real-time risk evaluation
- provide “safest route” alternatives
- incorporate transport mode risk exposure (walk, transit, car)

Goal: move from static analysis to decision-making assistance.

User Personalization

Introduce user-specific preferences such as:

- risk tolerance (e.g., avoid high-risk areas vs shortest path)
- preferred transport modes
- time-of-day sensitivity

Goal: adapt system outputs to individual user behavior.

Real-Time Data Ingestion

Transition from batch ETL pipelines to streaming ingestion:

- near real-time incident updates
- incremental feature updates
- event-driven processing

Goal: reduce latency between real-world events and system insights.

Spatial Grid Modeling (Advanced)

Leverage grid-based spatial modeling:

- divide the city into uniform cells
- compute localized risk metrics per cell
- complement clustering with continuous spatial coverage

Goal: enable fine-grained spatial intelligence beyond district-level analysis.

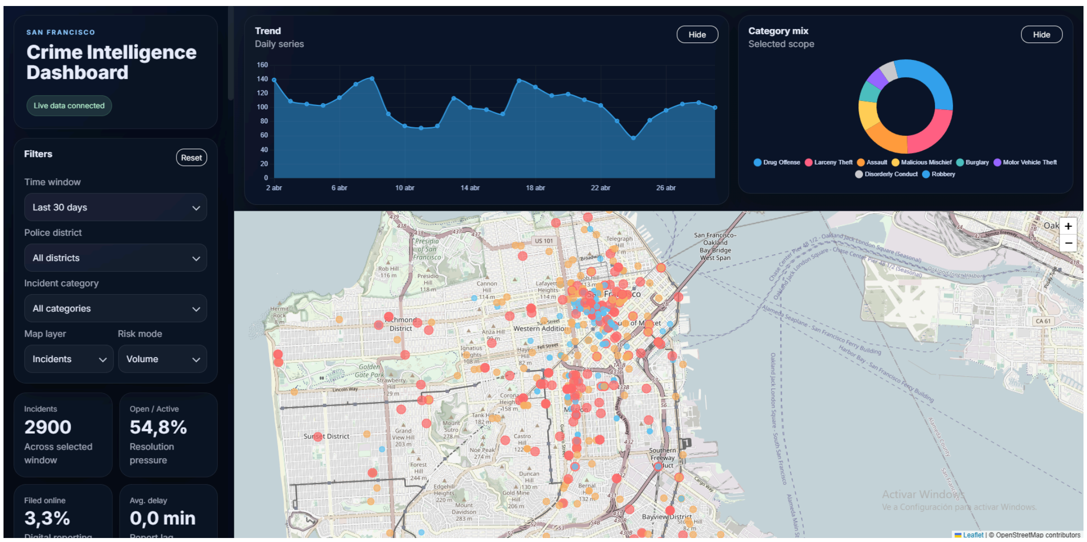
10. Demo / UI

Interactive Map

The map is the core of the application, providing multiple layers of analysis:

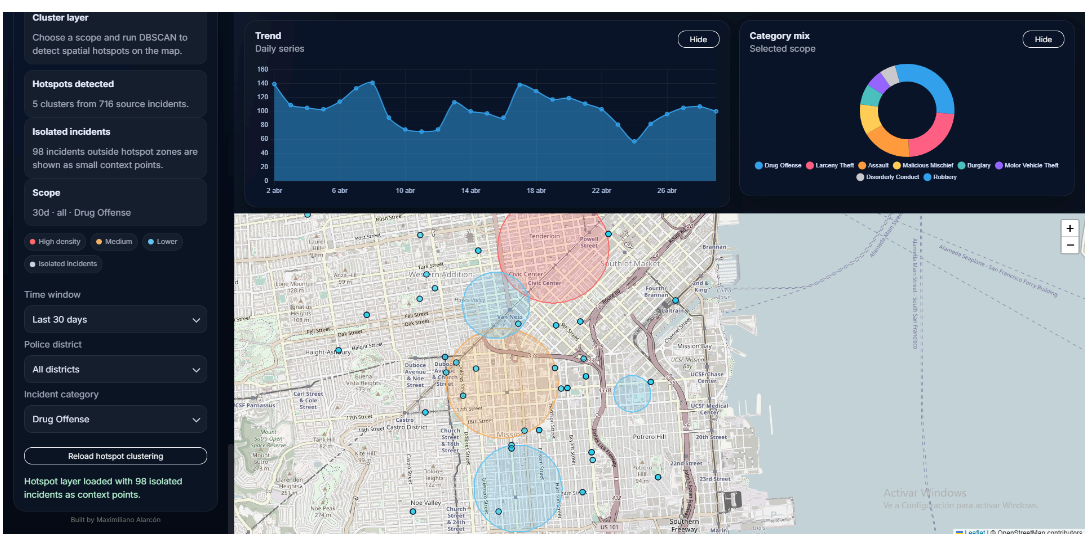
These layers can be toggled dynamically, allowing users to switch between different analytical perspectives.

Incident Points: Individual events displayed with dynamic styling based on risk signals

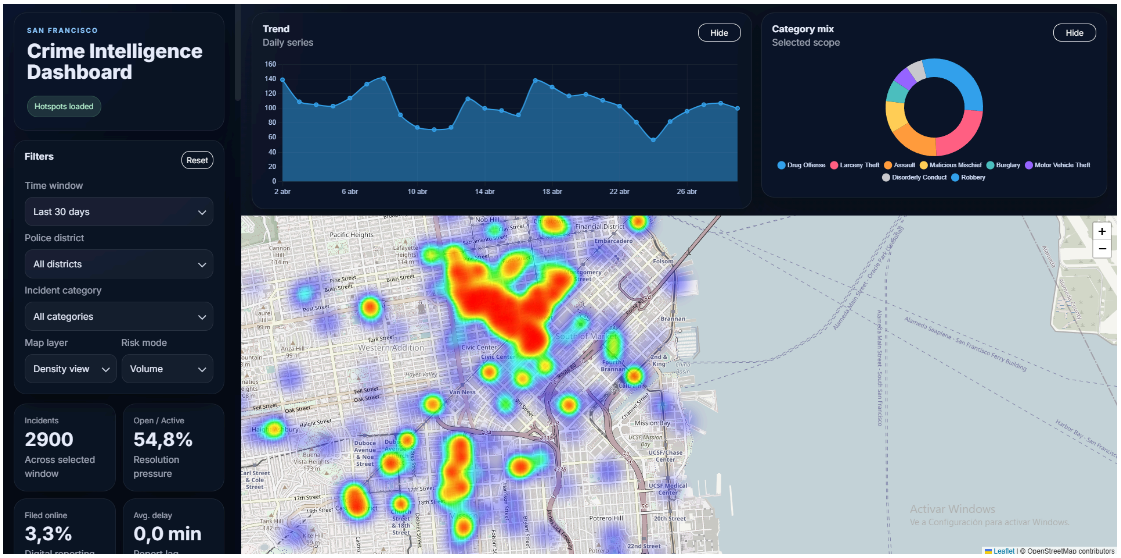


Hotspot Clusters: Density-based clusters generated using DBSCAN Includes both:

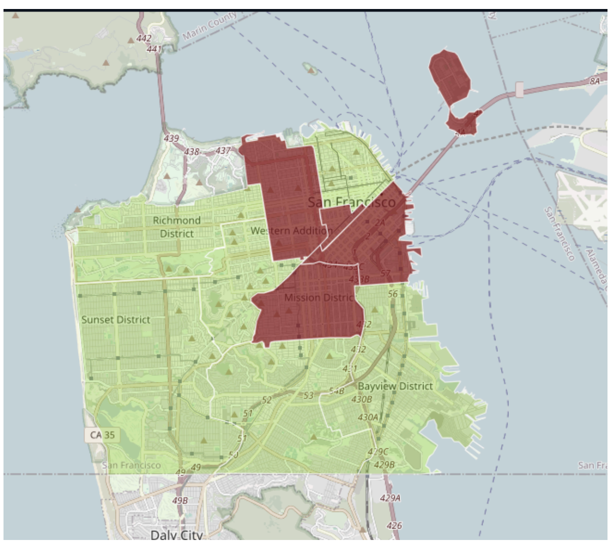
- high-density polygons
- isolated incidents (contextual points)

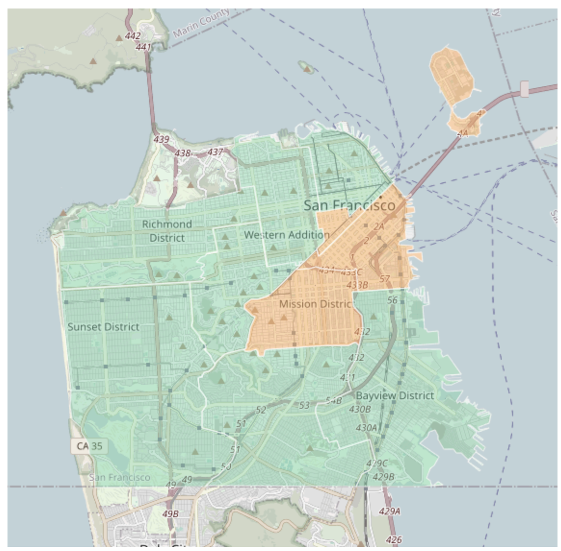


Heatmap Layer: Continuous density visualization for quick pattern recognition



Prediction Layers: Risk projections, Volume forecasts



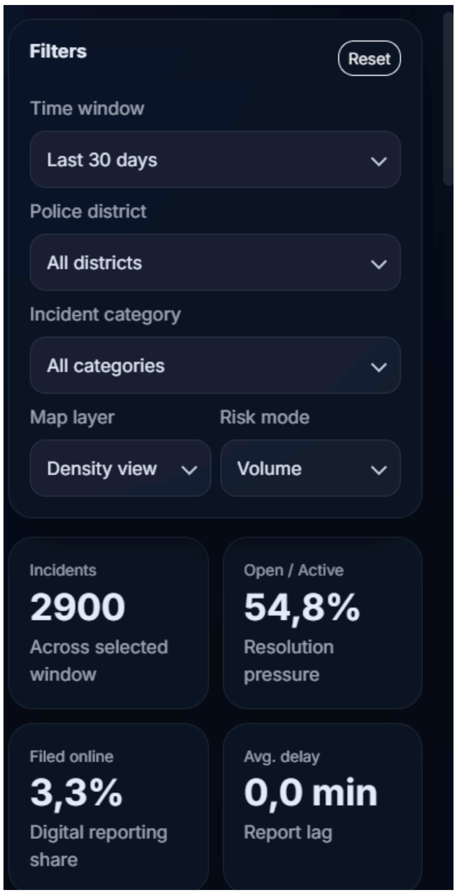


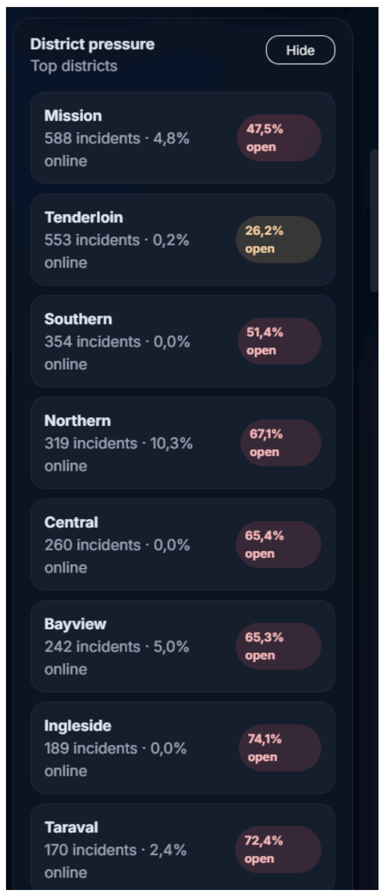
Dashboard & KPIs

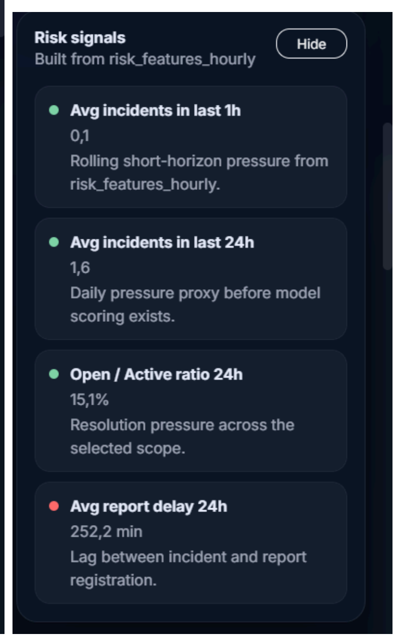
The dashboard provides a high-level overview of the current situation:

- Total incidents within the selected time window
- Open/active case ratio
- Online reporting percentage
- Aggregated risk signals

All metrics update in real time based on filters.







Users can refine the analysis using:

- Time window (24h, 7d, 30d)
- Police district
- Incident category

Filtering is applied consistently across:

- map layers
- charts

Analytical Views

Additional visual components support deeper exploration:

- Trend Charts
  Time-series evolution of incidents
- Category Distribution
  Breakdown of incidents by type
- District Pressure Ranking
  Comparative analysis across zones